

Module: Skladište (WMS — warehouse / pallet management) — DESIGN

Status: DESIGN / not built yet. This is the plan to approve before coding. Ported from the legacy standalone "Skladište" app v3.1 (see [Arhiva skripta/Skladište/3.1/SKLADISTE_KOMPLETNI_GUIDE.md](#)). We re-implement the proven domain model in our stack (SQLAlchemy 2.0 + Alembic + Jinja2/HTMX) and **fix the known legacy bugs by design** rather than copying the raw-sqlite code.

Scope

Pallet-level warehouse management for the label-printing plant: - **Zaprimanje** (receiving): create a plan, algorithm suggests positions, double-scan (pallet + position) to confirm, override allowed. - **Izdavanje** (issuing): FIFO/FEFO, bulk scan session with rollback. - **Inventura** (stocktake): scan, report missing / unexpected, reconcile. - **Prioriteti** (placement rules per šifra): clustering + 4 placement modes. - **Mapa skladišta** (2D/3D occupancy) — later phase. - **Naljepnice** (QR labels) — later phase.

v1 is fully self-contained: all pallet data is parsed from our own QR code (we generate the labels). No ERP lookup, no Materijali link, no user/login.

Out of scope for v1 (deferred): ERP (Pauk) barcode lookup, šifra→Materijali linking, any login / user tracking (all come later via the ERP), ngrok/mobile tunnel (our app already has LAN access via [dev-wifi.bat](#)).

Design decisions (these are the legacy fixes — non-negotiable)

1. **ISO dates everywhere in the DB** (`Date` / `DateTime` columns, not `dd.mm.yyyy` strings). Legacy substring date-parsing is bug B7/B8 → FIFO issued the wrong pallet. Format to Croatian only in the template layer (we already have date filters elsewhere).
2. **kolicina is numeric** (`Numeric` / `Float`), not TEXT → enables sums and partial issue.
3. **Position validation is mandatory** on every store/edit. Legacy had a validator it never called (bug B2/B17 → phantom pallets on non-existent positions). A position is valid iff it parses and resolves against the `REGALI` constant (see #12).
4. **Racks are a fixed code constant, never rewritten from the UI** (see #12) — kills the legacy code-injection bug B14 (`/api/warehouse/save` rewrote `warehouse.py` from input).
5. **One active pallet per position** enforced by a **partial unique index** (`UNIQUE(pozicija) WHERE datum_out IS NULL`). Kills the double-booking race (B3) and the resurrect-onto-occupied rollback bug (B6) at the DB level.
6. **Transactions** wrap the multi-row flows (prijem confirm, bulk izdaj, inventura close) so a mid-flow failure can't leave orphan rows (B4).
7. **Indexes** on `qr_raw`, `sifra`, `pozicija`, `datum_out` (legacy had none → full scans).
8. **Placement algorithm returns exactly n or signals a shortage** — never a silent partial list (legacy B9: a short list passed the `if not prijedlozi` check).

9. **Position codes are stored EXACTLY as printed on the QR — no zero-padding** (`A1P1V1` , `A13P15V5`), never auto-padded. The numeric parts are **variable width**, so anything that orders/ranges positions must sort by the **parsed integers** (`regal#`, `pozicija#`, `visina#`), never by the raw string (else `P10` sorts before `P2`). The validator and placement algorithm already operate on parsed ints, so this is free — it just must never regress to string comparison. (Legacy used padded `R1AP03V2` ; our real warehouse uses unpadded — see "Stvarni raspored".)
10. **No login and no user tracking in v1** (DECIDED): no `korisnik` field anywhere. Who-did- what arrives later via the ERP-wide auth module. No secrets/tokens hardcoded.
11. **All pallet data comes from the QR in v1** (DECIDED): receiving parses our own QR string (`qr_id|datum_ulaza|rok_trajanja|kolicina|sifra|lot`) → no ERP lookup, no Materijali link. Dates are **normalized to ISO on parse** (so FIFO is correct regardless of QR date format). ERP/Pauk lookup is a clearly separate future phase, not v1.
12. **Racks are a FIXED code constant, NOT a DB table** (DECIDED): the 20 racks never change at runtime; if they ever change, a developer edits the constant. So there is **no `regal` table and no rack-editing UI** — which also fully kills legacy bug B14 (UI rewriting code). The validator, capacity totals, and map all read from this constant.

Tables (SQLAlchemy 2.0)

Regali — fixed config in code (no DB table, no CRUD)

Racks are defined once as a constant in the module (`warehouse.py` / `config.py`):

```
# (naziv, zona, broj_pozicija, broj_visina) -- map layout coords added later if needed
REGALI = (
    [(f"A{i}", "A", 15, 5) for i in range(1, 14)] + # A1..A13
    [(f"B{i}", "B", 15, 4) for i in range(1, 6)] + # B1..B5
    [(f"C1", "C", 30, 4)] +
    [(f"D1", "D", 27, 4)]
)
# 20 racks, 1503 positions total
```

Position format: `<ZONA><REGAL#>P<POZICIJA>V<VISINA>` , unpadded — e.g. `A13P15V5` = zona A, regal 13, pozicija 15, visina 5. A position is valid iff it parses and falls within its rack's `broj_pozicija` / `broj_visina` (looked up in `REGALI`).

```
validator regex: ^([A-D])(\d{1,2})P(\d{1,2})V(\d)$
bounds (per zone, all racks in a zone share dims):
A → regal 1..13, pozicija 1..15, visina 1..5
B → regal 1..5, pozicija 1..15, visina 1..4
C → regal 1, pozicija 1..30, visina 1..4
D → regal 1, pozicija 1..27, visina 1..4
```

Implemented once in a `pozicija` helper, called on every pallet store/edit.

STVARNI RASPORED SKLADIŠTA (MJERODAVNO — IZ QR POPISA)

Source: the real position-code ranges for this warehouse (4 zones, all racks within a zone share dimensions).

Zona	Regali	Pozicija/regal	Visina	Mjesta/regal	Ukupno zona
A	A1–A13 (13)	15	5	75	975
B	B1–B5 (5)	15	4	60	300
C	C1 (1)	30	4	120	120
D	D1 (1)	27	4	108	108
Σ	20 regala				1503 mjesta

Zone A is the only one with 5 levels and is ~65% of capacity. C and D are single long racks (likely horizontal).
 Ranges: **A1P1V1–A13P15V5**, **B1P1V1–B5P15V4**, **C1P1V1–C1P30V4**, **D1P1V1–D1P27V4**. This table is the source for the **REGALI** constant above.

paleta — active + historical pallets

Field	Type	Notes
id	int, PK	
qr_raw	text, indexed	raw scanned QR string (the pipe format below)
qr_id	varchar, nullable	parsed pallet id
sifra	varchar, indexed	article code, parsed from QR (no Materijali link in v1)
lot	varchar, nullable	parsed from QR
kolicina	numeric, nullable	numeric , parsed from QR; supports partial issue
datum_ulaza	date, nullable	ISO (normalized on parse); production/entry date
rok_trajanja	date, nullable, indexed	ISO (normalized on parse); for FEFO
pozicija	varchar, indexed	validated against REGALI constant
datum_in	datetime	when stored
datum_out	datetime, nullable, indexed	NULL = active ; set = issued (soft delete)

All of **sifra/lot/kolicina/datum_ulaza/rok_trajanja/qr_id** are **parsed from the QR string** **qr_id|datum_ulaza|rok_trajanja|kolicina|sifra|lot**; dates normalized to ISO at parse time.

Partial unique index: `Index("uq_paleta_aktivna_pozicija", "pozicija", unique=True, sqlite_where=text("datum_out IS NULL"), postgresql_where=text("datum_out IS NULL"))`.

prijem — receiving plan header

id PK · **sifra** · **broj_paleta** int · **datum_plan** date · **status** (**aktivan** / **završen** / **odustao**). Relationship **stavke** → PrijemStavka.

prijem_stavka — receiving plan line

id PK · **prijem_id** FK→prijem.id, indexed · **redni_broj** int · **pozicija** (validated) · **qr_raw** nullable (filled on confirm) · **datum_potvrda** datetime nullable (NULL=unconfirmed).

inventura — stocktake header

id PK · **datum_pocetka** datetime · **datum_kraja** datetime nullable · **status** (**aktivan** / **završen** / **ponistena**). Relationship **stavke** → InventuraStavka.

inventura_stavka

id PK · **inventura_id** FK, indexed · **qr_raw** · **pozicija_skenirana** · **datum_skeniranja** .

prioritet — placement rule per šifra

Field	Type	Notes
id	int, PK	
sifra	varchar, indexed	
mod	varchar	standardno / puni_rupe / lijevi / strogo_lijevo (legacy overloaded one int — we split it out)
rack_ids	text, nullable	CSV of allowed rack names (v1; a join table is the clean later option)
napomena	text, nullable	
aktivan	bool, default true	

skladiste_event — audit log

id PK · **timestamp** datetime · **tip** · **poruka** · **detalji** nullable. (What happened + when; the "who" is added later with ERP auth. Could fold into an ERP-wide audit log.)

Placement algorithm (**predlozi_mjesta(n, sifra))**

Port the v3.1 logic (it is the valuable IP), with the B9 fix: 1. **Clustering first** (when **sifra** given): free positions on racks already holding the same šifra; if it yields all **n**, return. 2. **By mode** (**prioritet.mod**, default **standardno**): - **standardno**: consecutive (same height, same rack) then any; respect **rack_ids**. - **puni_rupe**: partially-filled racks first (consolidate), then empty. - **lijevi**: racks sorted by **min_x** ascending (toward the exit first). - **strogo_lijevo**: left half only (by X median), **no fallback** — but return value **explicitly signals shortage** ({found: k, of: n}), callers must check. Lower heights (V1) fill first (good for FIFO reach). Helpers: **_uzastopne**, **_bilo_koje**, **_trazi**. "Consecutive" = same height within one rack (no cross-rack spanning).

FIFO/FEFO issue

Sort candidate active pallets for a šifra by **datum_ulaza** (FIFO) or **rok_trajanja** (FEFO) — now trivial and correct because both are real **Date** columns (no string parsing). Warn when issuing a newer pallet while an older one exists. Bulk session: collect scans in a transaction, "Završi i spremi" (commit) or "Završi bez spremanja" (rollback all).

Routes (**/skladiste, HTMX partials per our convention)**

- GET /skladiste** — dashboard (metrics: zauzeto/slobodno/ukupno/ističe-30d) + tab nav (capacity totals come from the **REGALI** constant — no regali routes/CRUD)
- Zaprimanje: **GET /skladiste/zaprimanje**, **POST ... /plan** (calls **predlozi_mjesta**), **POST ... /potvrđi**, **POST ... /override**, **POST ... /zatvori**, **POST ... /odustani/{id}**, **GET**

... /plan/{id}/pdf

- Izdavanje: GET /skladiste/izdaj , GET ... /sifra/{sifra}?metoda=fifo|fefo , POST ... /izdaj , POST ... /izdaj/ponisti , GET ... /izdaj/pdf
- Pregled: GET /skladiste/pregled (paginated — legacy returned whole tables), POST ... /paleta/ispravak
- Rokovi: GET /skladiste/rokovi?dani=90 (+ pdf)
- Inventura: POST ... /inventura/start , GET ... /inventura/aktivna , POST ... /inventura/skeniraj , POST ... /inventura/zatvori , POST ... /inventura/ponisti , + reconciliation actions (otpis/usvoji — new, legacy lacked these)
- Prioriteti: GET/POST /skladiste/prioriteti , PUT/DELETE ... /{id}
- Mapa: GET /skladiste/mapa (zone overview), GET /skladiste/mapa/{zona} (zone grid, HTMX partial) — loads only the requested zone, never all 1503 at once
- Naljepnice (later): POST ... /naljepnice

Mapa skladišta (UI) — DECIDED: per-zone drill-down, CSS grid

Chosen for reliability + UX (1503 positions is too many to show readably at once). No Canvas/Plotly — plain Tailwind CSS-grid cells: themeable (respects dark mode, legacy 2D didn't), clickable (legacy canvas had no click handler), no build step, always rendered (legacy hid the map by default — bug B8).

Three levels: 1. **Pregled (/skladiste/mapa)** — 4 zone cards (A/B/C/D) with occupancy % + counts (e.g. **Zona A 612/975, 63%**), color-coded. Click a zone to drill in. 2. **Zona (/skladiste/mapa/{zona})** — each rack of the zone drawn as a **pozicija × visina** CSS grid (A7 = 15×5, C1 = 30×4 ...). Cell = one position. **V1 at the bottom** (ground level), pozicije left→right. Rows/cols laid out by **parsed ints** (not string order). Cell color by status: free / occupied / rok ≤30d / expired / search-highlight. 3. **Čelija** — hover = pallet tooltip (šifra, lot, rok, kolicina); click = detail + actions (izdaj / ispravak / premjesti).

Data: GET /skladiste/mapa/{zona} returns occupancy for that zone's racks only (one query filtered to active pallets in those positions). Search (by QR or šifra) highlights matching cells within the relevant zone. 3D view is explicitly **dropped** (not worth the Plotly cost; the 2D grid covers the need).

Templates (app/templates/skladiste/)

list.html /dashboard, _table_body.html , zaprimanje.html , izdaj.html , pregled.html , inventura.html , prioriteti.html , mapa.html (+ zone partial), partials per HTMX section — following CLAUDE.md conventions (slate card headers, HTMX search/partials, Alpine for scan UI). No **regali.html** (racks are a code constant).

Dual-mode architecture (standalone now + ERP module later)

Build **skladiste** as a **self-contained package**: - **Core** (palette, pozicije, placement algo, FIFO/FEFO, inventura, mapa, QR parse) is fully ERP-agnostic — **v1 needs nothing external; all data comes from the QR**. - Same package runs **standalone** (own **main.py** + own SQLite) OR mounts as a **module** in this ERP via **include_router(...)**. - A thin **ERP-adapter seam** is left for the future lookup (**ErpAdapter.lookup_barcode**), but it is **not built or needed in v1** — see the future ERP section below.

ERP integration — FUTURE (not v1; v1 reads everything from the QR)

Deferred. v1 is QR-only and self-contained. This section is the plan for when/if the ERP lookup is added later (`PaukAdapter` now, `NasErpAdapter` if our ERP takes off).

When added: the current-ERP (Pauk) API doesn't exist yet as a fixed contract — IT implements **to our spec** and generates Swagger afterward. Transport: **REST + HTTP Basic auth**, both apps on the **local network** (`https://pauk/...`). **Read-only** (standing "Pauk = samo čitanje" rule); write-back only on explicit approval.

Hard requirements we impose (to avoid the legacy bugs at the source): - All dates in the response as **ISO 8601** `yyyy-mm-dd` (no `dd.mm.yyyy` — kills FIFO bug B7). - `kolicina` as a **JSON number**, not a string.

Endpoint 1 — article/pallet lookup by barcode (primary):

```
GET /api/skladiste/artikl?barkod={barcode}      Authorization: Basic
200 → { "sifra": "12345", "naziv": "Etiketa X 100x50", "jedinica": "kom",
        "kolicina": 5000, "rok_tranja": "2027-03-15", "lot": "L2026-0042",
        "datum": "2026-06-20" }                # rok_tranja/datum ISO; kolicina numeric
404 → not found
```

Endpoint 2 — lookup by šifra (resolve/validate): `GET /api/skladiste/artikl/{sifra}` (same shape).

Endpoint 3 — write-back (DEFERRED, needs approval): `POST /api/skladiste/primka`, `POST /api/skladiste/izdavanje` — only if the read-only rule is lifted.

Our side stores ERP connection settings in an `erp_config` table (base URL, Basic creds, field mapping) + admin UI — never hardcoded (legacy hardcoded its ngrok token, B15).

Build phases (suggested)

1. **Foundation:** `REGALI` constant + position validator + QR parser (→ ISO dates, numeric qty); models + Alembic migration (FK, indexes, partial-unique active position) for paleta/prijem/inventura/prioritet/event; module skeleton + dashboard (capacity from `REGALI`). No `regali` table/CRUD, no auth, no ERP.
2. **Core flow:** zaprimanje (scan QR → parse → `predlozi_mjesta`) → FIFO/FEFO izdaj → inventura (+ reconcile).
3. **Prioriteti** + placement modes.
4. **Mapa** — per-zone drill-down, Tailwind CSS-grid (see "Mapa skladišta (UI)" above). No Canvas/Plotly, no 3D.
5. **Naljepnice** (QR PDF — we already use reportlab in reklamacije; this is what produces the QR strings v1 parses).
6. **(Future) ERP lookup** — `PaukAdapter` + `erp_config` + REST contract. Only if/when the ERP path is pursued; not part of the core build.

Dependencies / open questions

- **Auth:** DECIDED — no login and no user tracking in v1 (no `korisnik` anywhere). Comes later via ERP-wide auth. Admin-gating is moot (racks aren't editable; prioriteti stays open).

- **Šifra ↔ Materijal:** DECIDED — **no link in v1** (unnecessary without the ERP). Pallet `sifra` is just text parsed from the QR. May connect to `materijal` later alongside the ERP.
 - **Map fidelity:** DECIDED — per-zone drill-down, Tailwind CSS-grid, no Canvas/Plotly, no 3D (see "Mapa skladišta (UI)").
 - **Barcode source:** v1 keeps our own QR pipe format; ERP (Pauk) barcode lookup is phase 6 via `PaukAdapter` (REST + Basic, read-only, ISO dates — see "ERP integration").
 - **Deployment topology:** DECIDED — warehouse app runs on the **local server (LAN)**, same network as Pauk; it calls `https://pauk/ ...` over LAN with Basic auth (no public exposure).
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Generirano iz docs/12-modul-skladiste.md - ERP/MES/WMS